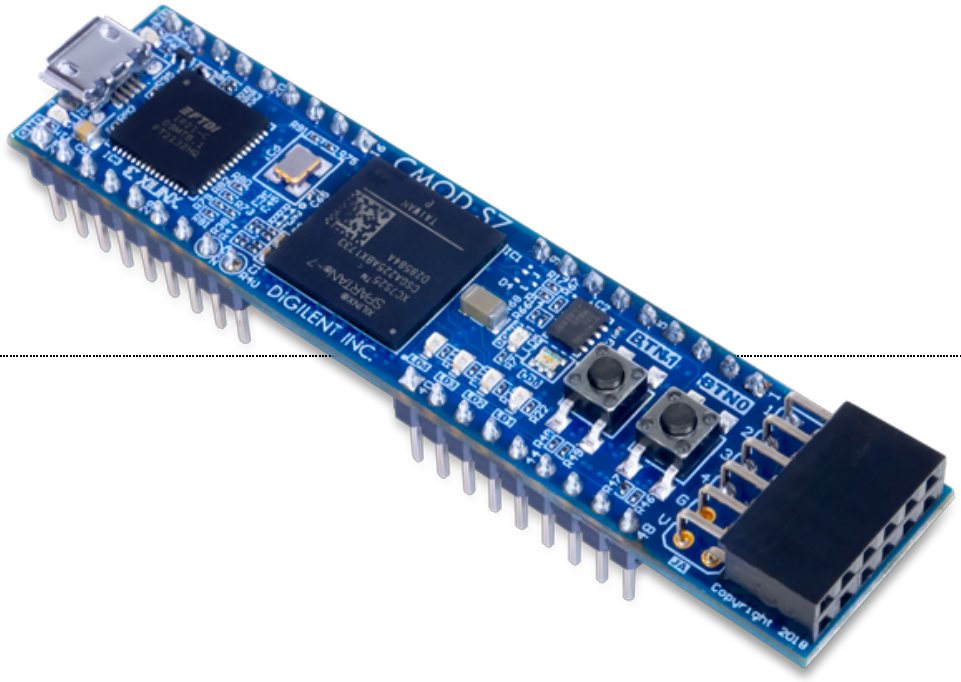
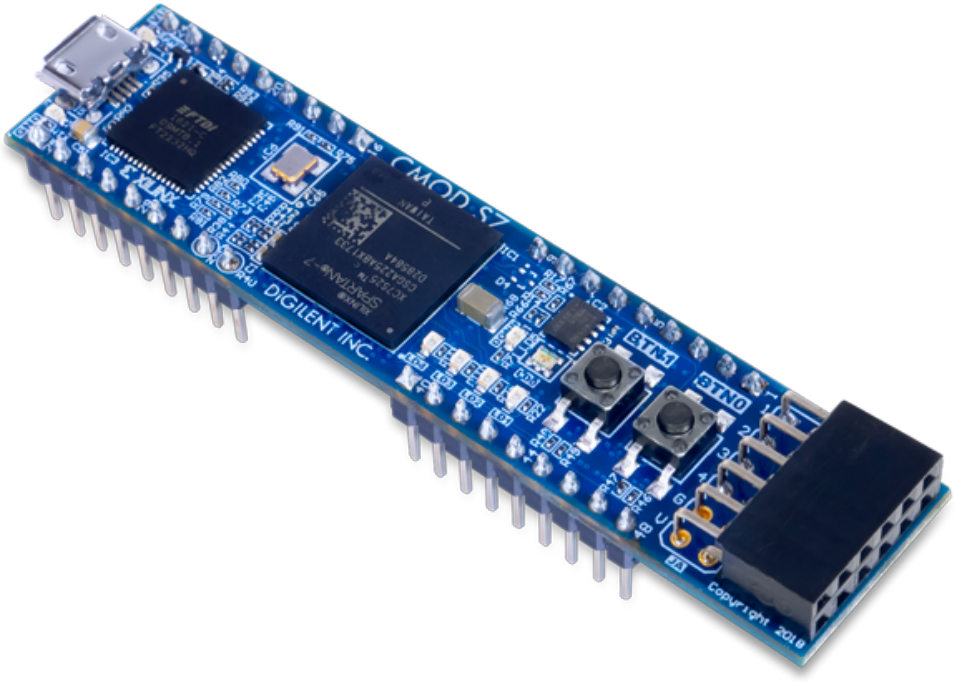
Cmod S7 Reference Manual

The Digilent Cmod S7 is a small, 48-pin DIP form factor board built around a Xilinx Spartan 7 FPGA. 32 FPGA digital I/O signals, 2 FPGA analog input signals, an external power input rail, and ground are routed to 100-mil-spaced through-hole pins, making the Cmod S7 well suited for use with solderless breadboards. At just 0.7" by 3.05" inches, it can be loaded in a standard socket and used in embedded systems. The board also includes a programming ROM (), clock source, USB programming and data transfer circuit, power supplies, LEDs, and buttons.

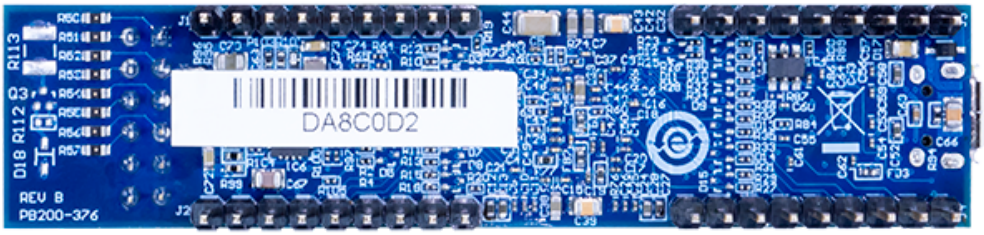


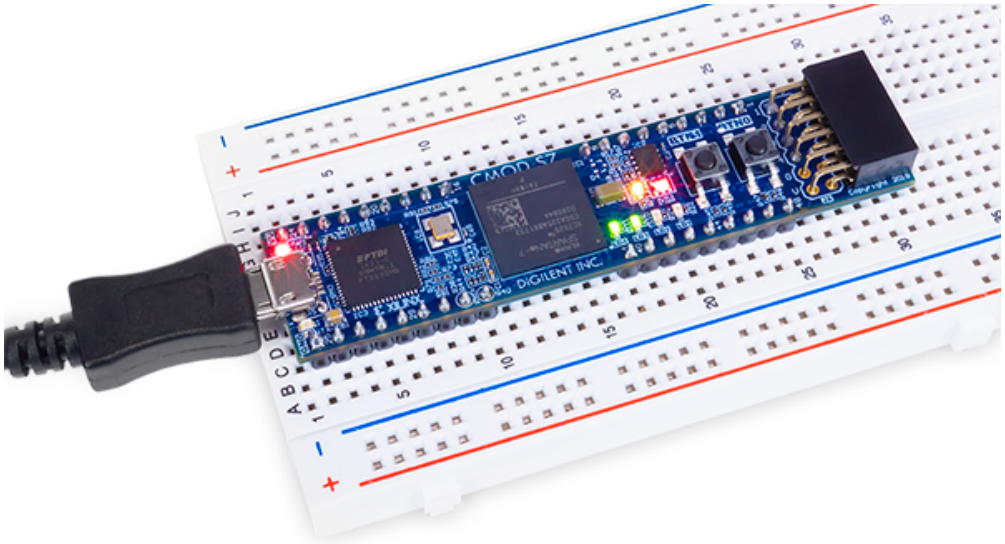
(https://digilent.com/reference/_media/reference/programmable-logic/cmod-s7/cmod_s7-obl-600.png)





Feedback

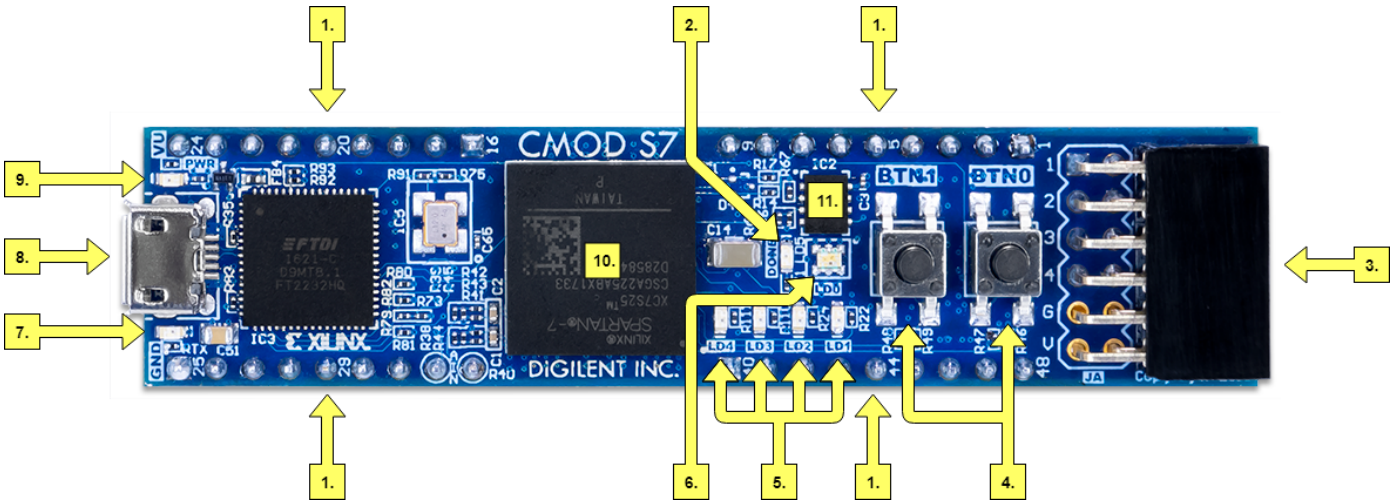




Features

- **Xilinx Spartan-7 FPGA (XC7S25-1CSGA225C)**
  - 3,650 slices containing four 6-input LUTs and 8 flip-flops
  - 1,620 Kbits of fast block RAM ()
  - 3 clock management tiles, each with a phase-locked loop and mixed-mode clock manager
  - 80 DSP slices
  - Internal clock speeds exceeding 450 MHz ()
  - On-chip analog-to-digital converter (XADC)
  - Programmable over JTAG and Quad-SPI Flash
- **Memory**
  - 4 MB () Quad-SPI Flash
- **Power**
  - Powered from USB or 5V external supply connected to DIP pin 24
- **USB**
  - USB-JTAG programming circuitry
  - USB-UART bridge
- **Push-buttons and LEDs**
  - 2 Buttons
  - 4 LEDs
  - 1 RGB LED ()
- **Expansion Connectors**
  - 1 Pmod connector
    - 8 total FPGA I/O
  - 48-pin DIP form-factor header
    - 32 total FPGA I/O
    - 2 single-ended 0-3.3V analog inputs to XADC
    - 2 power pins

Feedback



(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-callout.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Callout	Description	Callout	Description
1	48-pin DIP form factor header	7	UART status LED ()
2	FPGA programming DONE LED ()	8	Shared USB JTAG/UART port
3	Pmod connector	9	Power good LED ()
4	User push buttons	10	Spartan-7 FPGA
5	User LEDs	11	SPI Flash
6	User tri-color LED ()		

Software Support

The Cmod S7 is fully compatible with the high-performance Vivado ® Design Suite versions 2018.1 and newer. It is supported under the free WebPACK™ installation option, which does not require a license, so designs can be implemented at no additional cost. This free installation includes the ability to create MicroBlaze™ soft-core processor designs. Design resources, example projects, and tutorials are available for download at the Cmod S7 Resource Center (https://digilent.com/reference/programmable-logic/cmod-s7/start).

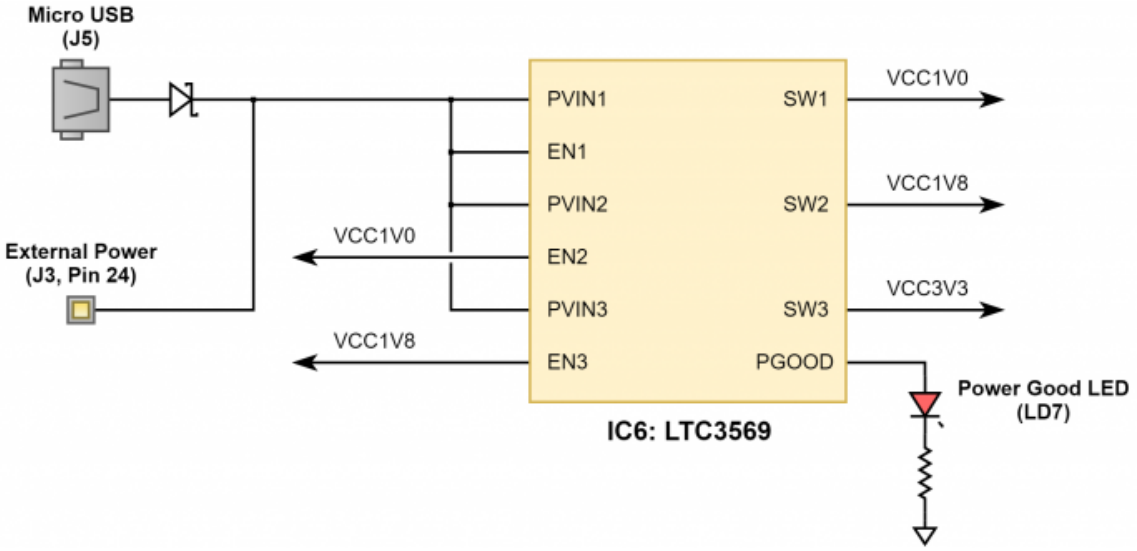
The Cmod S7 can be used with Vivado 2017.4, but WebPACK installations of this version of Vivado may not contain the device files for the specific Spartan-7 FPGA used on the Cmod S7. See this Xilinx Answer Record (https://support.xilinx.com/s/article/70505) for more information, and for a solution.

Functional Description

1 Power Supplies

The Cmod S7 uses a Linear Technologies LTC3569 triple output buck power regulator to generate the needed voltages for the onboard components. It is connected as shown in Figure 1.1. The characteristics of the outputs are shown in Table 1.1.

Feedback



(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-power.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 1.1 Power Supply

Supply	Circuits	Device	Maximum Current
3.3V	FPGA User I/O, USB Controller, Flash Memory, Pmod Connector, LEDs, Buttons	IC6: LTC3569	1.2A
1.8V	FPGA, USB Controller	IC6: LTC3569	0.6A
1.0V	FPGA Core and Block RAM ()	IC6: LTC3569	0.6A

Table 1.1 Power Rail Characteristics

1.1 Power Input Options

It is possible to power the Cmod S7 from either USB or from an external power supply attached to pins 24 and 25 of the DIP headers, labeled "VU" and "GND ()", respectively. The required characteristics of a power source attached to either of these two options is outlined in Table 1.1.1.

Connector Type	Connector Label	Min/Rec/Max Voltage
USB	J5	Per USB Specification
DIP	J3 (Pin 24)	4.5/5.0/5.5

When the Cmod S7 is powered by the USB connector, the voltage from the USB device is driven onto the VU pin. This makes it possible to power an external circuit from the USB host in addition to the Cmod S7. The VU pin is driven via a schottky diode, so a small voltage drop will occur, but it should be assumed that the voltage present on VU will be close to 5V.

Warning: When a USB host is attached to the micro USB connector, the VU pin on the DIP headers (pin 24) is driven to the voltage being provided by the USB host (typically 4.5V-5.5V). To avoid risk of damage, any power source attached to the VU pin must be disconnected before a USB host is attached. Not taking this precaution can be particularly dangerous if the power source is a battery.

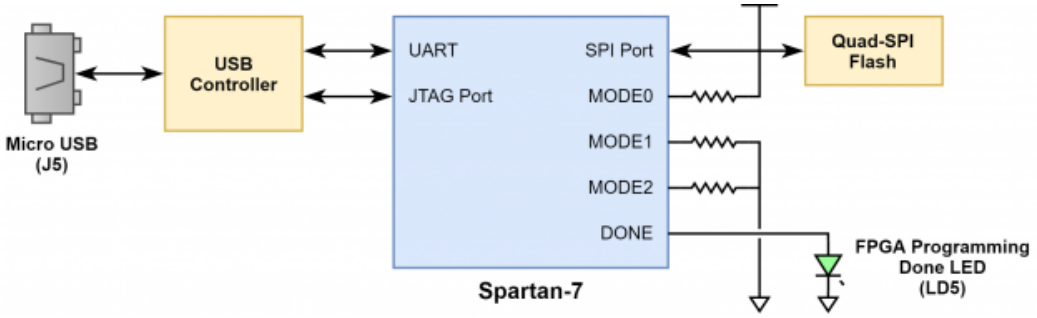
2 FPGA Configuration

After power-on, the Spartan-7 FPGA must be configured (or programmed) before it can perform any functions. The FPGA can be configured in one of two ways:

- A PC can use the Digilent USB-JTAG circuitry (port J5) to program the FPGA any time the power is on.
- A file stored in the nonvolatile serial (SPI) flash device can be transferred to the FPGA using the SPI port.

Figure 2.1 shows the different options available for configuring the FPGA.

Feedback



(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-config.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 2.1 Configuration

The FPGA configuration data is stored in files called bitstreams that have the .bit file extension. The Vivado software from Xilinx can create bitstreams from VHDL, Verilog®, or block-level designs.

Bitstreams are stored in volatile memory cells within the FPGA. This data defines the FPGA's logic functions and circuit connections, and it remains valid until it is erased by removing board power, or by writing a new configuration file using the JTAG port.

A Spartan-7 25T bitstream is typically 9,935,224 bits. The time it takes to program the Cmod S7 can be decreased by compressing the bitstream before programming, and then allowing the FPGA to decompress the bitstream itself during configuration. Depending on design complexity, compression ratios of 10x can be achieved. Bitstream compression can be enabled within the Xilinx tools to occur during generation, and is enabled by default in Digilent's master XDC file for the Cmod S7. For more information on how to do this, consult the Xilinx documentation for the toolset being used.

After successful programming, the FPGA will cause the "DONE" LED () to illuminate.

The following sections provide greater detail about programming the Cmod S7 using the different methods available.

2.1 JTAG Configuration

The Xilinx tools typically communicate with FPGAs using the Test Access Port and Boundary-Scan Architecture, commonly referred to as JTAG. During JTAG programming, a .bit file is transferred from the PC to the FPGA using the onboard Digilent USB-JTAG circuitry (port J5). JTAG programming can be performed any time after the Cmod S7 has been powered on. If the FPGA is already configured, then the existing configuration is overwritten with the bitstream being transferred over JTAG.

Programming the Cmod S7 with an uncompressed bitstream using the onboard USB-JTAG circuitry usually takes around 6 seconds. JTAG programming can be performed by Vivado's Hardware Manager.

2.2 Quad-SPI Configuration

Since the FPGA's memory on the Cmod S7 is volatile, it relies on the Quad-SPI flash memory to store the configuration between power cycles. This configuration mode is called Master SPI. The blank FPGA takes the role of master and reads the configuration file out of the flash device upon power-up. To that effect, a configuration file needs to first be written to the flash. When programming a non-volatile flash device, a bitstream file is transferred to the flash in a two-step process. First, the FPGA is programmed with a circuit that can program flash devices, and then data is transferred to the flash device via the FPGA circuit (this complexity is hidden from the user by the Xilinx tools). This is called indirect programming. After the flash device has been programmed, it will automatically configure the FPGA at any subsequent power-on event. Programming files stored in the flash device will remain until they are overwritten, regardless of power-cycle events.

Quad-SPI programming can be performed using Vivado's Hardware Manager.

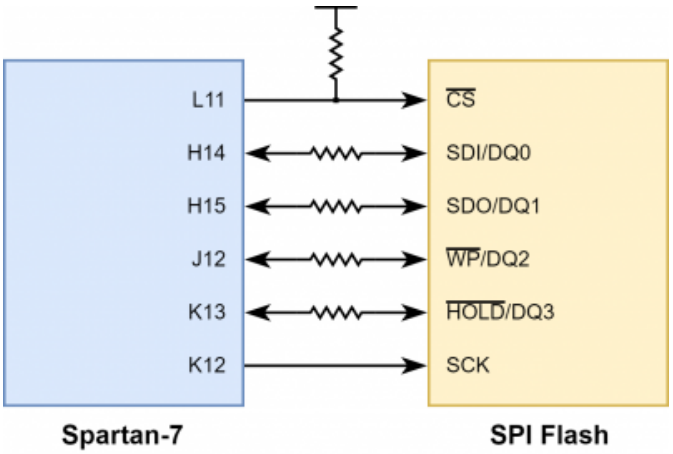
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3 Quad-SPI Flash

FPGA Configuration files can be written to the Quad-SPI Flash (Macronix part number MX25L3233F). The FPGA will automatically read a configuration file from this device at power on. A Spartan-7 25T configuration file requires 9,935,224 bits of memory, leaving about 69% of the flash device (or ~2.8 MB ()) available for user data. A common use for this extra memory is to store MicroBlaze programs too big to fit in the onboard block memory. These programs are then loaded and executed using a smaller bootloader program that can fit in the block memory. It is possible to automatically generate this bootloader, roll it into the bitstream, and then program the bitstream and large MicroBlaze program into t Quad SPI Flash using Xilinx SDK.

The contents of the memory can be manipulated by issuing certain commands on the SPI bus. The implementation of this protocol is outside th scope of this document. Xilinx's AXI Quad SPI IP core can be used to read/write the flash in a MicroBlaze design. Refer to Xilinx's product guide (https://docs.xilinx.com/v/u/en-US/pg153-axi-quad-spi) for this core to learn more about using it, or to Macronix's datasheet (http://www.macronix.com/Lists/Datasheet/Attachments/6744/MX25L3233F,%203V,%2032Mb,%20v1.6.pdf) for the flash device to learn how to implement a custom controller. All signals in the SPI bus are general-purpose user I/O pins after FPGA configuration and can be used like any other FPGA I/O.

Feedback



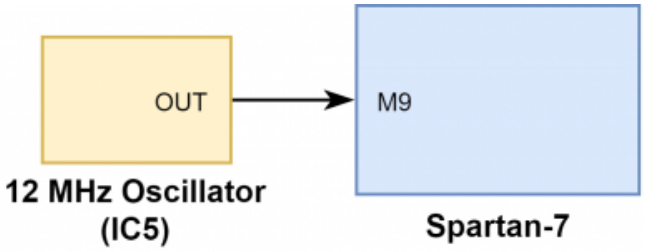
(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-flash.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 3.1 Flash Interface

4 Oscillators/Clocks

The Cmod S7 includes a 12 MHz () crystal oscillator connected to pin M9 (an MRCC input on bank 14). This clock is intended to be used as a general purpose system clock. The clock can drive MMCMs to generate clocks of various frequencies and with known phase relationships that may be needed throughout a design. The 12 MHz () input clock cannot directly drive a PLL because they have a minimum input frequency of 19 MHz (). Some rules restrict which MMCMs and PLLs may be driven by the 12 MHz () input clock. For a full description of these rules and of the capabilities of the Spartan-7 clocking resources, refer to Xilinx UG472 (https://docs.xilinx.com/v/u/en-US/ug472_7Series_Clocking), titled "7 Series FPGAs Clocking Resources User Guide".

Xilinx offers the Clocking Wizard IP core to help users generate the different clocks required for a specific design. This wizard will properly instantiate the needed MMCMs and PLLs based on the desired frequencies and phase relationships specified by the user. The wizard will then output an easy-to-use wrapper component around these clocking resources that can be inserted into the user's design. The clocking wizard can be accessed from within the Vivado and IP Integrator tools.



(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-clocking.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 4.1 Clock Input

5 USB-UART Bridge (Serial Port)

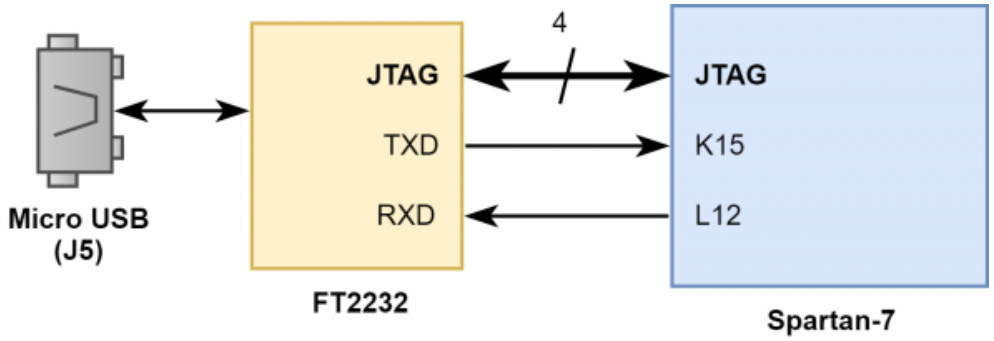
The Cmod S7 includes an FTDI FT2232HQ USB-UART bridge (attached to Micro-USB connector J5) that allows the user to use PC applications to communicate with the board using standard Windows COM port commands. Free USB-COM drivers, available from FTDI's website (https://www.ftdichip.com) under the "Virtual Com Port" or VCP heading, convert USB packets to UART/serial data. Serial port data is exchanged with the FPGA using a two-wire serial port (TXD/RXD). After the drivers are installed, I/O commands can be used from the PC directed to the COM port to produce serial data traffic on the K15 and L12 FPGA pins.

An onboard status LED () (LD6) provides visual feedback on traffic flowing through the port.

The FT2232HQ is also used as the controller for the Digilent USB-JTAG circuitry, but the USB-UART and USB-JTAG functions behave netirely independant of one another. Programmers interested in using the UART functionality of the FT2232 within their designs do not need to worry about the JTAG circuitry interfering with the UART data transfers, and vice-versa. The combination of these two features into a single device allows the Cmod S7 to be programmed, communicated with via UART, and powered from a computer attached with a single Micro USB cable.

The connections between the FT2232HQ and the Spartan-7 are shown in Figure 5.1. Signal names that imply direction are from the point-of-view of the DTE (Data Terminal Equipment), in this case, a PC connected to the Micro USB port.

Feedback



(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-usbuart.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 5.1 USB-UART Bridge

6 Basic I/O

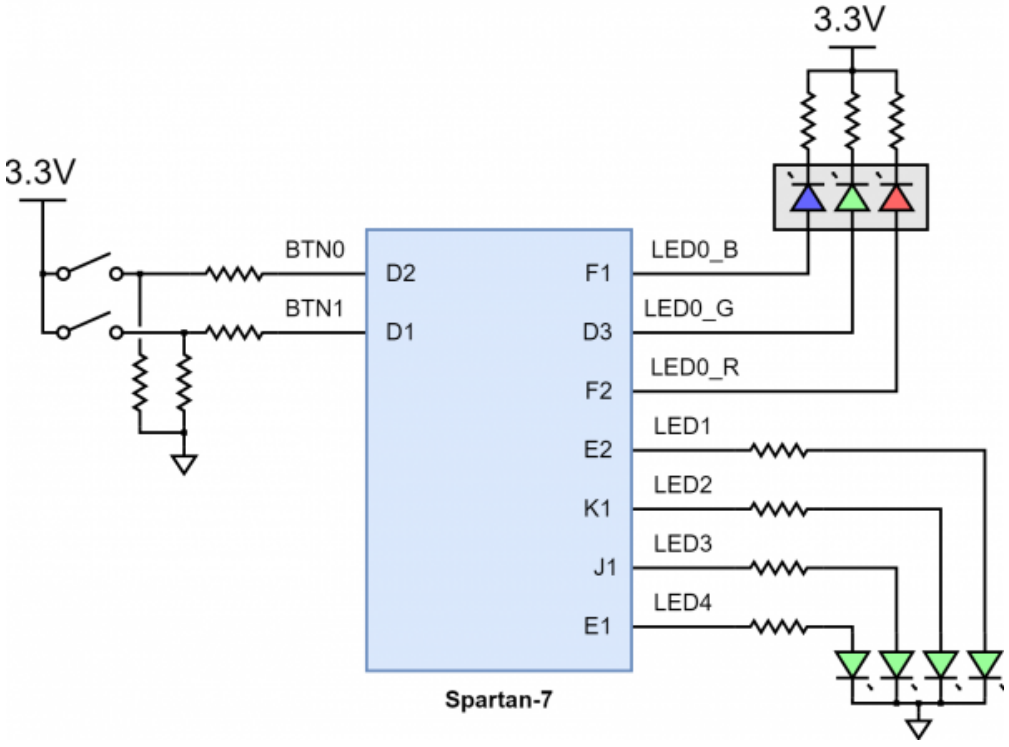
The Cmod S7 includes one RGB LED (), 4 individual LEDs, and 2 push buttons, as shown in Figure 6.1. The push buttons are connected to the FPGA via series resistors to prevent damage from inadvertent short circuits (a short circuit could occur if an FPGA pin assigned to a button was inadvertently defined as an output). The buttons are "momentary" switches that normally generate a low output when they are at rest, and a high output only when they are pressed.

The four individual high-efficiency LEDs are anode-connected to the FPGA via 330 Ohm resistors, so that they will turn on when a logic-high voltage is applied to their respective I/O pins. Additional LEDs that are not user-accessible indicate power-on, FPGA programming status, and USB-UART traffic.

The single RGB LED () has three input signals that drive the cathodes of three smaller internal LEDs: one red, one blue, and one green. Driving the signal corresponding to one of these colors low will illuminate the internal LED (). The RGB LED () will emit a color dependent on the combination of internal LEDs that are currently being illuminated. For example, if the red and blue signals are driven low and green is driven high, the RGB LED () will emit a purple color.

Note: Digilent strongly recommends the use of pulse-width modulation (PWM) when driving the RGB LED (). Driving any of the signals to a steady logic '0' will result in the LED () being illuminated at an uncomfortably bright level. This can be avoided by ensuring that none of the RGB signals are driven with more than a 50% duty cycle. Using PWM also greatly expands the potential color palette of the RGB LED (). Individually adjusting the duty cycle of each each color between 50% and 0% causes the different colors to be illuminated at different intensities, allowing virtually any color to be displayed.

Figure 6.1 shows how each of the LEDs and push buttons are connected to the Spartan-7.

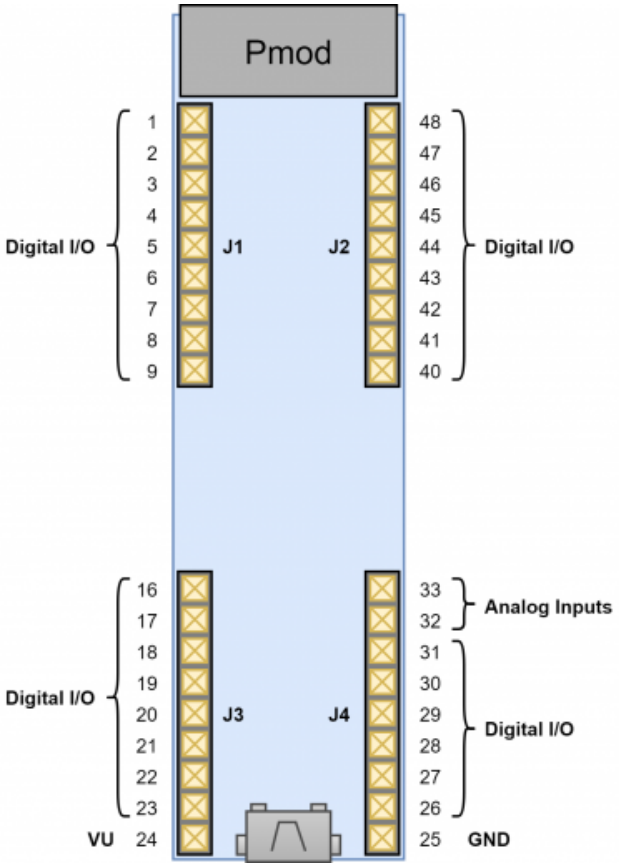


(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-basic-io.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 6.1 Cmod S7 Basic I/O

7 DIP Header

The Cmod S7 has a 48-pin DIP form factor connector, implemented as four 9-pin headers for connecting to breadboards and custom fixtures. The pins have 100 mil spacing, and the entire module is 0.7 inches by 3.05 inches. Headers J1 and J3 are separated by 700 mil lengthwise along the Cmod, measured from the center of the innermost pins, as are headers J2 and J4. Headers J1 and J2 are separated by 600 mil across the board, as are headers J3 and J4. Of the 48 possible pin locations, 36 are populated on the four headers. Of these, 32 are directly connected to FPGA Digital I/Os, 2 are voltage-divided and connected to FPGA analog inputs, and two are connected to power pins. The pin and header numbers are shown in Figure 7.1.



(https://digilent.com/reference/_detail/reference/programmable-logic/cmod-s7/cmod-s7-dip.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 7.1 DIP Pin Diagram

VU (pin 24) can be used to power the Cmod S7 when it is not connected to a USB Host. When the device is connected to a USB host, VU is driven by the 5V rail of the USB connector and can be used to power devices external to the Cmod S7. See the Power Supplies section for more information.

7.1 DIP Digital I/O

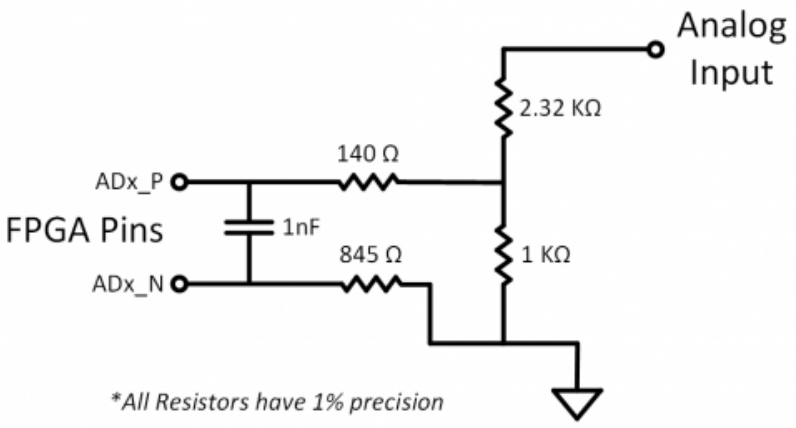
The pins directly connected to the FPGA can be used as general purpose inputs or outputs. Each of these pins is connected to the FPGA via a 240 Ohm series resistor. The series resistors prevent short circuits that can occur if a user accidentally drives a signal that is supposed to be used as an input. The downside to this added protection is that these resistors limit the maximum switching speed of these signals to 25 MHz (). These pins support input voltages of up to 3.5V.

For more information on the electrical characteristics of these pins, please see Xilinx's Spartan-7 Datasheet (https://docs.xilinx.com/v/u/en-US/ds189-spartan-7-data-sheet).

7.2 Analog Inputs

Pins 32 and 33 of the DIP headers are used as analog inputs to the XADC module of the FPGA. The FPGA expects that the inputs range from 0V-1V, so resistor-divider circuits are used on the Cmod S7 to scale down the input voltage from 3.3V. This circuit is shown in Figure 7.2.1. This circuit allows the XADC module to accurately measure any voltage between 0V and 3.3V (relative to GND () on pin 25) that is applied to these pins. The analog inputs can tolerate input voltages of up to 5.5V, though any voltage above 3.3V will be read as 3.3V by the XADC module.

Feedback



(https://digilent.com/reference/_media/cmod_a7/cmoda7_b_analog.png)

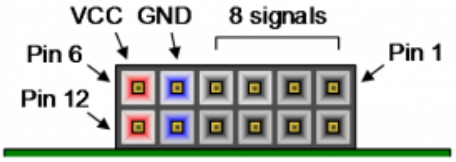
Figure 7.2.1 Analog Input Circuit

The XADC core within the Spartan-7 is a dual-channel 12-bit analog-to-digital converter capable of operating at 1 MSPS. Either channel can be driven by either of the two auxiliary analog inputs to the DIP pins. The XADC core is controlled and accessed from a user design via the Dynamic Reconfiguration port (DRP). The DRP also provides access to a temperature sensor that is internal to the FPGA. For more information on using the XADC core, refer to Xilinx UG480 (https://docs.xilinx.com/v/u/en-US/ug480_7Series_XADC), titled "7 Series FPGAs and Zynq-7000 All Programmable SoC XADC Dual 12-Bit 1 MSPS Analog-to-Digital Converter". A demo that uses the XADC core is available on the Cmod S7 Resource Center (https://digilent.com/reference/programmable-logic/cmod-s7/start)

8 Pmod Connector

Pmod connectors are 2×6, right-angle, 100-mil spaced female connectors that mate with standard 2×6 pin headers. Each 12-pin Pmod connector provides two 3.3V VCC () signals (pins 6 and 12), two Ground signals (pins 5 and 11), and eight logic signals, as sown in Figure 8.1. The VCC () and Ground pins can deliver up to 200mA of current, but care must be taken not to exceed any of the power budgets of the onboard regulator or the external power supply (described in the Power Supplies section).

Warning: *Since the Pmod pins are connected to Spartan-7 FPGA pins using a 3.3V logic standard, care should be taken not to drive these pins over 3.4V.*



(https://digilent.com/reference/_detail/basys3-pmod_connector.png?id=programmable-logic%3Acmod-s7%3Areference-manual)

Figure 8.1 Pmod Diagram

Digilent provides a large collection of Pmod accessory boards that can attach to Pmod expansion connectors to add ready-made functions like A/D ()'s, D/A ()'s, motor drivers, sensors, and other functions. See www.digilentinc.com (http://www.digilentinc.com) for more information.

The Cmod S7 has one "standard" type Pmod connector, and the FPGA pin connections for it are described in Table 8.1. Standard Pmod connectors are connected to the FPGA via 200 Ohm series resistors. The series resistors prevent short circuits that can occur if a user accidentally drives a signal that is supposed to be used as an input. The downside to this added protection is that these resistors can limit the maximum switching speed of the data signals.

	Pmod JA
Pmod Type	**Standard**
Pin 1	J2
Pin 2	H2
Pin 3	H4
Pin 4	F3
Pin 7	H3
Pin 8	H1
Pin 9	G1

Feedback

	Pmod JA
Pin 10	F4

Table 8.1. Cmod S7 Pmod Pinout

cmod-s7 (https://digilent.com/reference/tag/cmod-s7?do=showtag&tag=cmod-s7), doc (https://digilent.com/reference/tag/doc?do=showtag&tag=doc)

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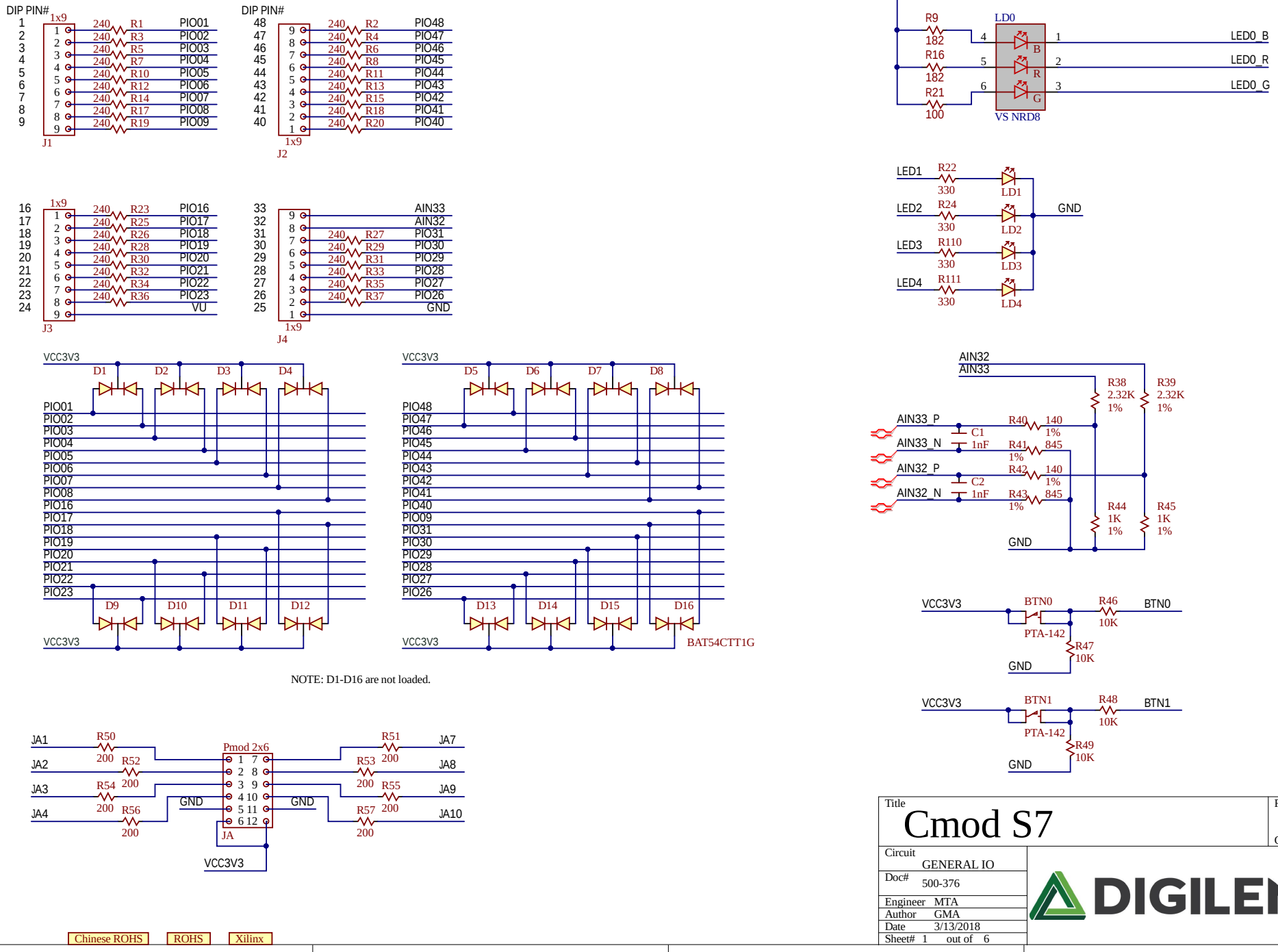
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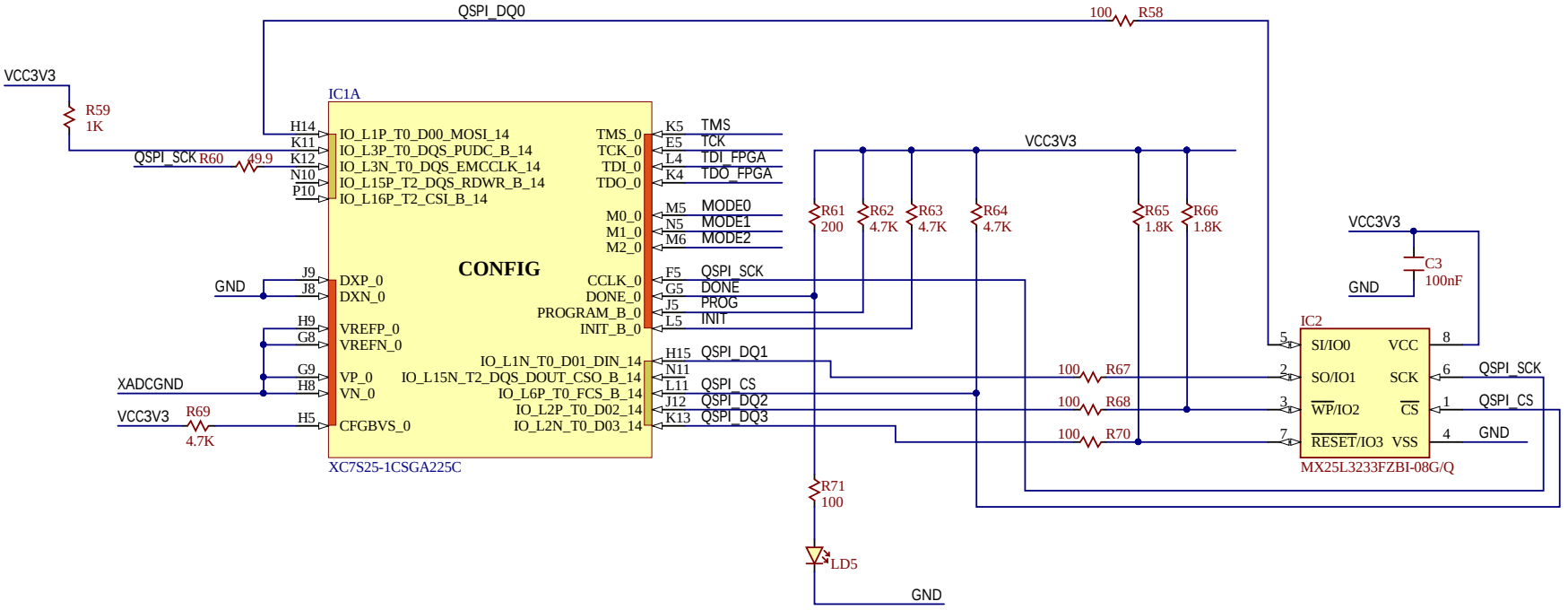


NOTE: D1-D16 are not loaded.

Chinese ROHS ROHS Xilinx

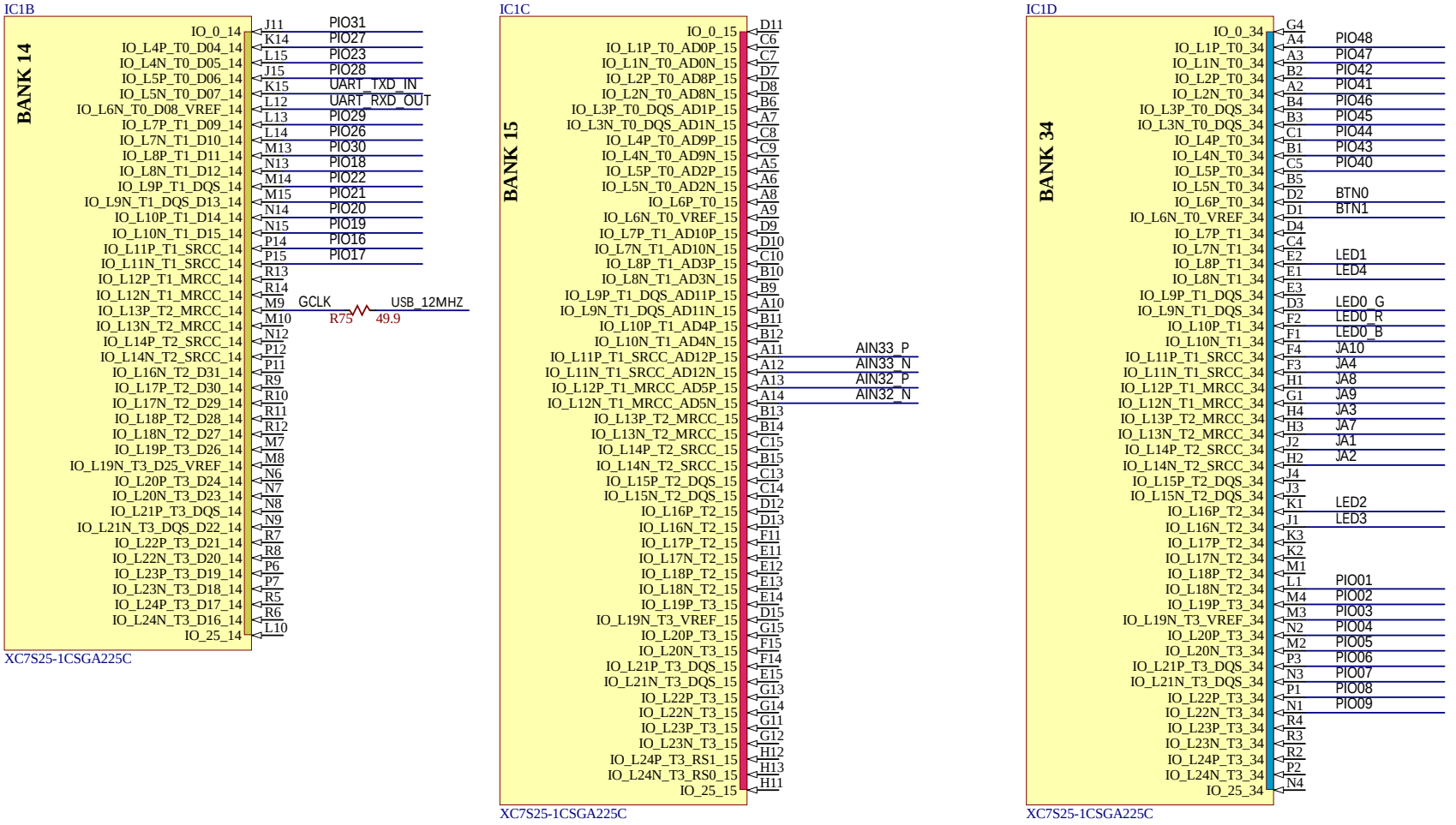
Title	Cmod S7
Rev	B.0
Copyright	2018
Circuit	GENERAL IO
Doc#	500-376
Engineer	MTA
Author	GMA
Date	3/13/2018
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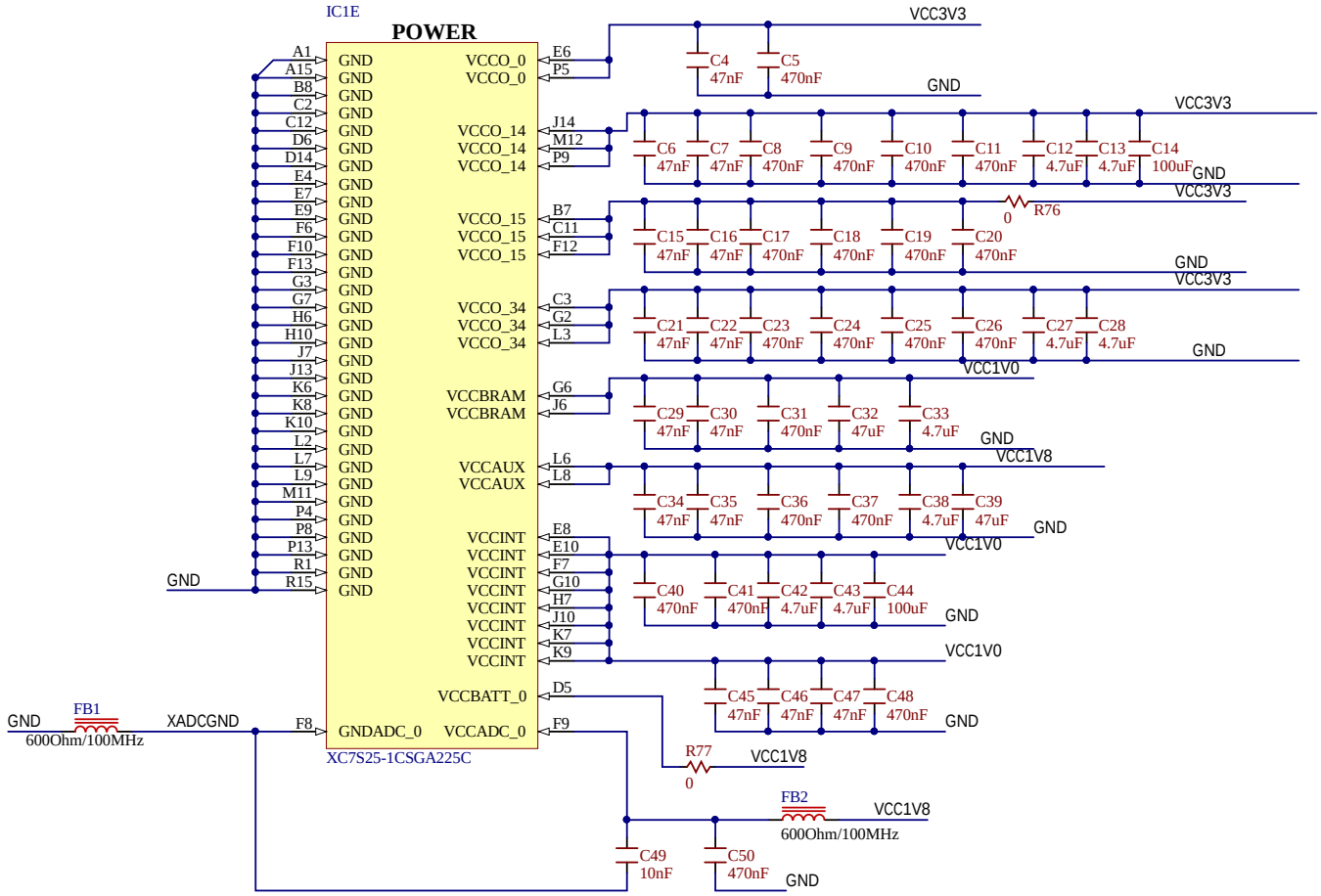
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Circuit	CONFIG, SPI FLASH
Doc#	500-376
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IC1B
BANK 14
IO_0_14
IO_L4P_T0_D04_14
IO_L4N_T0_D05_14
IO_L5P_T0_D06_14
IO_L5N_T0_D07_14
IO_L6N_T0_D08_VREF_14
IO_L7P_T1_D09_14
IO_L7N_T1_D10_14
IO_L8P_T1_D11_14
IO_L8N_T1_D12_14
IO_L9P_T1_DQS_14
IO_L9N_T1_DQS_D13_14
IO_L10P_T1_D14_14
IO_L10N_T1_D15_14
IO_L11P_T1_SRCC_14
IO_L11N_T1_SRCC_14
IO_L12P_T1_MRCC_14
IO_L12N_T1_MRCC_14
IO_L13P_T2_MRCC_14
IO_L13N_T2_MRCC_14
IO_L14P_T2_SRCC_14
IO_L14N_T2_SRCC_14
IO_L16N_T2_D31_14
IO_L17P_T2_D30_14
IO_L17N_T2_D29_14
IO_L18P_T2_D28_14
IO_L18N_T2_D27_14
IO_L19P_T3_D26_14
IO_L19N_T3_D25_VREF_14
IO_L20P_T3_D24_14
IO_L20N_T3_D23_14
IO_L21P_T3_DQS_14
IO_L21N_T3_DQS_D22_14
IO_L22P_T3_D21_14
IO_L22N_T3_D20_14
IO_L23P_T3_D19_14
IO_L23N_T3_D18_14
IO_L24P_T3_D17_14
IO_L24N_T3_D16_14
IO_25_14
J11
K14
L15
J15
K15
L12
L13
L14
M13
N13
M14
M15
N14
N15
P14
P15
R13
R14
M9
M10
N12
P12
P11
R9
R10
R11
R12
M7
M8
N6
N7
N8
N9
R7
R8
P6
P7
R5
R6
L10
PIO31
PIO27
PIO23
PIO28
UART_TXD_IN
UART_RXD_OUT
PIO29
PIO26
PIO30
PIO18
PIO22
PIO21
PIO20
PIO19
PIO16
PIO17
GCLK
R75
49.9
USB_12MHZ
XC7S25-1CSGA225C
IC1C
BANK 15
IO_0_15
IO_L1P_T0_AD0P_15
IO_L1N_T0_AD0N_15
IO_L2P_T0_AD8P_15
IO_L2N_T0_AD8N_15
IO_L3P_T0_DQS_AD1P_15
IO_L3N_T0_DQS_AD1N_15
IO_L4P_T0_AD9P_15
IO_L4N_T0_AD9N_15
IO_L5P_T0_AD2P_15
IO_L5N_T0_AD2N_15
IO_L6P_T0_15
IO_L6N_T0_VREF_15
IO_L7P_T1_AD10P_15
IO_L7N_T1_AD10N_15
IO_L8P_T1_AD3P_15
IO_L8N_T1_AD3N_15
IO_L9P_T1_DQS_AD11P_15
IO_L9N_T1_DQS_AD11N_15
IO_L10P_T1_AD4P_15
IO_L10N_T1_AD4N_15
IO_L11P_T1_SRCC_AD12P_15
IO_L11N_T1_SRCC_AD12N_15
IO_L12P_T1_MRCC_AD5P_15
IO_L12N_T1_MRCC_AD5N_15
IO_L13P_T2_MRCC_15
IO_L13N_T2_MRCC_15
IO_L14P_T2_SRCC_15
IO_L14N_T2_SRCC_15
IO_L15P_T2_DQS_15
IO_L15N_T2_DQS_15
IO_L16P_T2_15
IO_L16N_T2_15
IO_L17P_T2_15
IO_L17N_T2_15
IO_L18P_T2_15
IO_L18N_T2_15
IO_L19P_T3_15
IO_L19N_T3_VREF_15
IO_L20P_T3_15
IO_L20N_T3_15
IO_L21P_T3_DQS_15
IO_L21N_T3_DQS_15
IO_L22P_T3_15
IO_L22N_T3_15
IO_L23P_T3_15
IO_L23N_T3_15
IO_L24P_T3_RS1_15
IO_L24N_T3_RS0_15
IO_25_15
D11
C6
C7
D7
D8
B6
A7
C8
C9
A5
A6
A8
A9
D9
D10
C10
B10
B9
A10
B11
B12
A11
A12
A13
A14
B13
B14
C15
B15
C13
C14
D12
D13
F11
E11
E12
E13
E14
D15
G15
F15
F14
E15
G13
G14
G11
G12
H12
H13
H11
AIN33_P
AIN33_N
AIN32_P
AIN32_N
XC7S25-1CSGA225C
IC1D
BANK 34
IO_0_34
IO_L1P_T0_34
IO_L1N_T0_34
IO_L2P_T0_34
IO_L2N_T0_34
IO_L3P_T0_DQS_34
IO_L3N_T0_DQS_34
IO_L4P_T0_34
IO_L4N_T0_34
IO_L5P_T0_34
IO_L5N_T0_34
IO_L6P_T0_34
IO_L6N_T0_VREF_34
IO_L7P_T1_34
IO_L7N_T1_34
IO_L8P_T1_34
IO_L8N_T1_34
IO_L9P_T1_DQS_34
IO_L9N_T1_DQS_34
IO_L10P_T1_34
IO_L10N_T1_34
IO_L11P_T1_SRCC_34
IO_L11N_T1_SRCC_34
IO_L12P_T1_MRCC_34
IO_L12N_T1_MRCC_34
IO_L13P_T2_MRCC_34
IO_L13N_T2_MRCC_34
IO_L14P_T2_SRCC_34
IO_L14N_T2_SRCC_34
IO_L15P_T2_DQS_34
IO_L15N_T2_DQS_34
IO_L16P_T2_34
IO_L16N_T2_34
IO_L17P_T2_34
IO_L17N_T2_34
IO_L18P_T2_34
IO_L18N_T2_34
IO_L19P_T3_34
IO_L19N_T3_VREF_34
IO_L20P_T3_34
IO_L20N_T3_34
IO_L21P_T3_DQS_34
IO_L21N_T3_DQS_34
IO_L22P_T3_34
IO_L22N_T3_34
IO_L23P_T3_34
IO_L23N_T3_34
IO_L24P_T3_34
IO_L24N_T3_34
IO_25_34
G4
A4
A3
B2
A2
B4
B3
C1
B1
C5
B5
D2
D1
D4
C4
E2
E1
E3
D3
F2
F1
F4
F3
H1
G1
H4
H3
J2
H2
J4
J3
K1
J1
K3
K2
M1
L1
M4
M3
N2
M2
P3
N3
P1
N1
R4
R3
R2
P2
N4
PIO48
PIO47
PIO42
PIO41
PIO46
PIO45
PIO44
PIO43
PIO40
BTN0
BTN1
LED1
LED4
LED0_G
LED0_R
LED0_B
JA10
JA4
JA8
JA9
JA3
JA7
JA1
JA2
LED2
LED3
PIO01
PIO02
PIO03
PIO04
PIO05
PIO06
PIO07
PIO08
PIO09
XC7S25-1CSGA225C
Title
Cmod S7
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Circuit
FPGA BANKS
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Engineer MTA
Author GMA
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Rev	B.0
Copyright	2018
Circuit	FPGA POWER
Doc#	500-376
Engineer	MTA
Author	GMA
Date	3/13/2018
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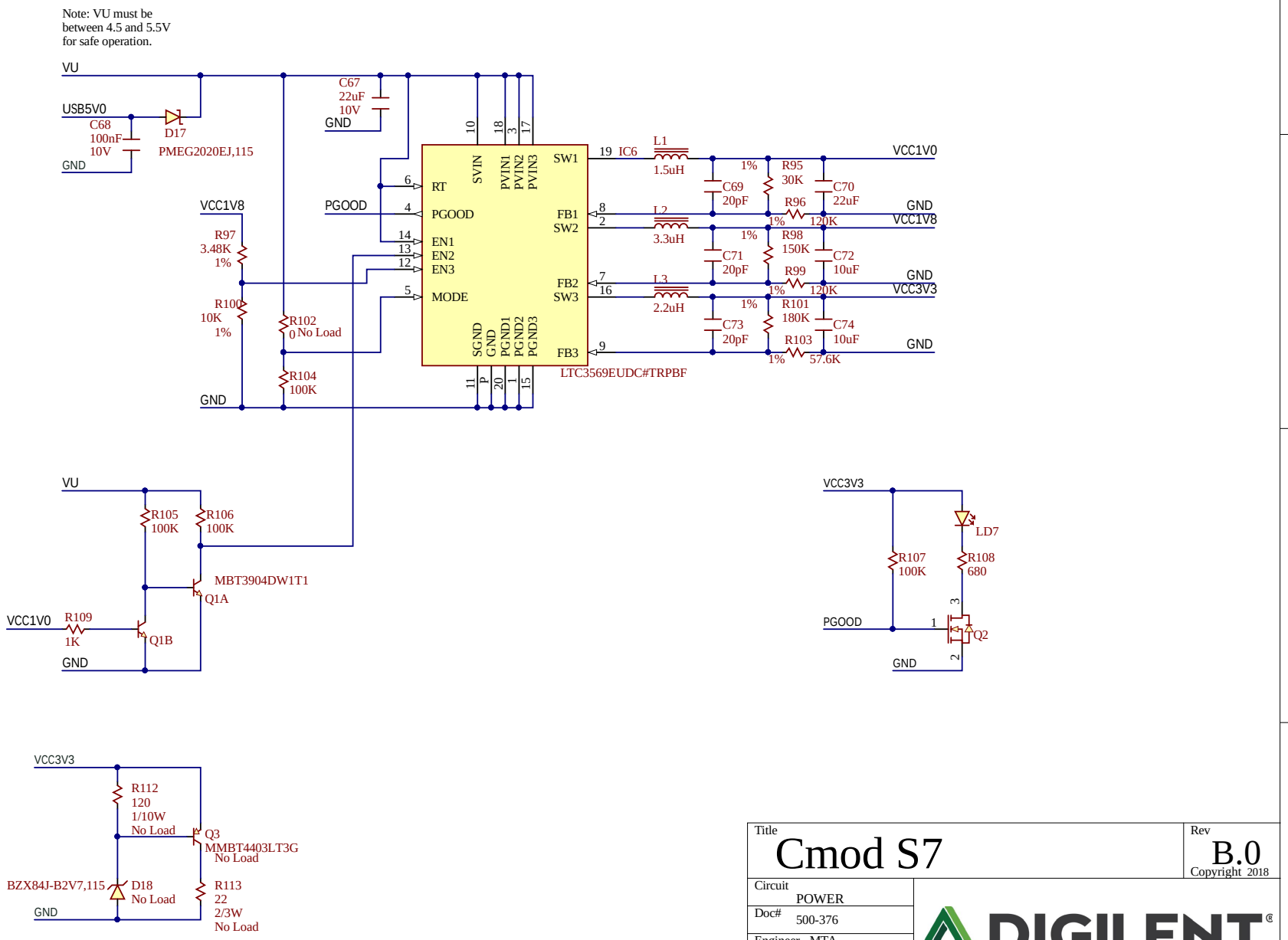
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Rev	B.0
Circuit	USB PROG/UART
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Engineer	MTA
Author	GMA
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Note: VU must be between 4.5 and 5.5V for safe operation.
VU
USB5V0
C68 100nF 10V
GND
D17
PMEG2020EJ,115
C67 22uF 10V
GND
VCC1V8
R97 3.48K 1%
R100 10K 1%
R102 0 No Load
R104 100K
GND
PGOOD
RT
PGOOD
EN1
EN2
EN3
MODE
SVIN
PVIN1
PVIN2
PVIN3
SW1
FB1
SW2
FB2
SW3
FB3
SGND
GND
PGND1
PGND2
PGND3
LTC3569EUDC#TRPBF
IC6
L1 1.5uH
L2 3.3uH
L3 2.2uH
C69 20pF
C71 20pF
C73 20pF
R95 30K 1%
R96 120K 1%
R98 150K 1%
R99 120K 1%
R101 180K 1%
R103 57.6K 1%
C70 22uF
C72 10uF
C74 10uF
VCC1V0
GND
VCC1V8
GND
VCC3V3
GND
VU
R105 100K
R106 100K
MBT3904DW1T1
Q1A
VCC1V0
R109 1K
Q1B
GND
VCC3V3
LD7
R107 100K
R108 680
PGOOD
Q2
GND
VCC3V3
R112 120 1/10W No Load
Q3 MMBT4403LT3G No Load
BZX84J-B2V7,115
D18 No Load
R113 22 2/3W No Load
GND
Title
Cmod S7
Rev
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Circuit
POWER
Doc#
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Engineer MTA
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